

The Impact of ICTS and Artificial Intelligence: How are they reshaping power structures, both international and domestic

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Abstract

With the development of Information and Communication Technologies (ICTs), Artificial Intelligence (AI), and Big Data technologies, domestic and international politics have inevitably been affected by technological advancements. This paper analyzes the restructuring of power structures by ICTs and AI, and conducts case studies. Our research will include both domestic and international aspects.

1 Main Concepts

1.1 Political Marketization

Public choice theory, a theory that uses economic tools to address traditional political issues, posits that politics inherently possesses market characteristics. Politicians gain votes by catering to voters' needs, voters influence policy through their votes, and interest groups influence legislation through lobbying. This aligns with the rational man assumption in economics, emphasizing the simultaneous existence of "self-interest" in politics and economics.

We need to clarify that marketization here does not mean a communist country adopting a market economy, but rather a phenomenon of incorporating market logic and economic rationality into the political process. At its core lies the "market-based" allocation and exchange of political resources such as power, votes, and policies. In some cases, this phenomenon also implies corruption. Furthermore, we are not referring to a neutral or reform-oriented understanding of the term here, but rather to marketization as a means of modern state governance—introducing market mechanisms to improve efficiency in specific areas (such as public service outsourcing and market access).

Finally, the issues we are discussing here do not refer to the traditionally existing problem of political marketization, but specifically to the political marketization brought about and promoted by information technology. We are particularly referring to situations where politics is influenced by technology, resulting in market-like political behavior and, in turn, competition for technology.

1.2 Data Sovereignty

In the era of newspapers and television, sovereignty was "physical." But in the era of big data and AI, sovereignty has become "computational."

Due to the marketization of politics and the influence of data and algorithm controllers, a country inevitably worries about foreign political influence if data and algorithms are controlled by foreigners. This concern has led the West, previously enthusiastic about promoting technology and standards, to discuss supply chain security, including not only physical but also digital aspects. Perhaps this is why the US is forcing China to sell TikTok. Both the Republican and Democratic parties, deeply aware of the power of digital technology in the election, agree that national digital sovereignty must be protected. We can see the intense competition between China and the US in AI-related fields.

1.3 Techno-protectionism

When traditional global markets are no longer advantageous to certain major powers, they leverage their remaining advantages in key areas (such as high-end chips, operating systems, and financial

clearing systems) to "weaponize" these technological nodes. This behavior forcibly severs the flow of global markets, forcing other countries (such as China) to build their own "full industrial chains" or "regional economic zones" for survival. In other words, technological protectionism, and the accompanying trade protectionism, leads to regionalization. This is not the original intention of technology, but rather a deliberate technological segmentation by leading powers in the great power game, using their remaining technological advantages to counter challengers. It is essentially a form of ideology aimed at protecting digital sovereignty, and also a form of defensiveness.

2 How do ICTs and AI reshape international competition among major powers and their domestic impact?

2.1 Phase1: From newspaper to television

As early as 1985, in *Amusing Ourselves to Death*, Neil Postman told us that what might become a reality is not the 1984-style future we fear, but rather the future predicted by Huxley in *Brave New World*.

Postman first tells us that the media defines what is "truth." In the print era, logic, order, objectivity, and depth were the standards of truth because reading required prolonged focus and rational thinking[Bus]. In the television era, however, visual pleasure, emotional impulses, and immediacy have become the standards of truth. If a politician doesn't appear "good" or humorous on television, even the most correct political views will struggle to gain acceptance.

These two technological shifts also represent a transformation in information and communication technology, from the "Age of Exposition" to the "Age of Show Business." New technologies quickly unified the world, and in this new world, politics became increasingly decontextualized and fragmented. One moment you might be reporting on a genocide in a certain region, and the next you'd be watching a mobile advertisement. In this fragmented environment, people don't need coherent logic, nor can they establish it. To adapt to the fragmentation of television, politics had to simplify complex issues, simplifying them to a single sentence. Politics no longer sells policy objectives but rather sells emotions and sensory stimulation, or even just a slogan—just look at that universally known slogan: MAGA!

Of course, even a visionary like Postman couldn't have fully predicted the technology and computing power we possess today, and how they would be applied to politics—a field in itself. Three scholars discussed this area in detail[HLT23]. As a practical communication technology, it involves algorithms, automation, and manual filtering to distribute content, potentially involving mobile networks or social media. This is far more targeted than television, which can only create a relatively unified public opinion sphere. Under the influence of IT, we can use big data to precisely target advertising and create individual information cocoons for each person. This far exceeds the efficiency of television—efficiency Postman in 1985 could never have imagined. Those who control information technology are no longer merely exploiting its fragmented nature but are actively using these tools to manipulate the content everyone might see. The marketization of politics seems to have transcended mere commercial competition and begun to become malignant.

At the international level, diplomacy between nations is also influenced by technological advancements. In the newspaper era, information transmission was slow and costly, and politics followed an elite diplomatic model, much like the domestic situation, largely excluding the general public. Diplomacy was a rational game played in private by a small group of highly educated elites, with the masses excluded from decision-making. Then came television, a tool clearly applicable to domestic politics, now broadcast globally via wireless communication. International politics thus entered the era of public diplomacy. This represents the marketization of international politics; nations not only speak to each other's governments but also showcase their "brand" to their people. International competition is no longer merely a comparison of tank and missile numbers but a contest of who can tell the best story. We witnessed this during the Cold War, with the US and the Soviet Union constantly demonstrating which system truly represents the light of humanity.

During this phase, television technology triggered the famous "CNN Effect".[Rob05] The live broadcasts of famine and war can quickly ignite public sentiment and force governments around the world to respond. At this moment, countries have seen the influence of technological progress, the importance of mastering technology, and the importance of the cultural soft power that is built on technology:

how can a young person who wears American jeans, listens to American music, and watches American movies hate America?

2.2 Phase2: From television to big data

Is that the end? Not at all, because we are quickly entering the AI era. As one of the most popular and impactful information technologies on human society today, large language models (sometimes simply referred to as AI), represented by ChatGPT, have had an unprecedented impact on humanity. If you don't know ChatGPT, you're definitely out of touch. Of course, AI technologies that can treat politics like business competition and employ any means necessary extend far beyond large language models. For example, during the 2016 US presidential election, a meme about Hillary Clinton circulated among far-right users, and was even retweeted by Trump[WH18]. Deception in the AI era is far more complex than simply "using BBC headlines or Times New Roman fonts to make them look like news," as Philip Howard mentioned[HLT23]. With the increasing maturity of AI technology, it becomes increasingly difficult for people to judge the authenticity of videos. Platform controllers use methods such as buying traffic on social media and funding data scientist teams to precisely target ads to voters, sometimes even intentionally including false information about competitors.

If this is the case in domestic politics, then in international competition, the underhanded tactics are even more pronounced. Even in the television era, a country could use film and television to propagate its message to the people of other nations. In the age of AI and big data, digital sovereignty is becoming more crucial than ever. Sometimes, even unseen by the public, covert political propaganda can occur. We know that the ideology of large-scale models is derived from their training materials, and scientists have found that chatting with large-scale models that pre-set viewpoints can significantly alter users' political orientations.[LCL⁺25][HTH⁺25] Furthermore, thanks to modern computing power and big data technology, whoever controls technological power controls this precise one-to-one retail market.

Digital sovereignty is so important that nations, especially those represented by the United States, are now forced to utilize their remaining technological advantages as a form of protection when facing challenges from emerging powers. They no longer emphasize globalization; their primary goal is instead to maintain their existing advantages, attempting to preserve their dominance in areas such as chips, AI technology, and quantum computing. This has created a trend of deglobalization, which is not the original intention of technology, but rather a microcosm of the struggle for digital sovereignty and technological protectionism.

3 Case Study

3.1 American Politics

Next, we will conduct a case study of the United States. In the 2016 US presidential election, Trump's goal from the outset was to control media attention, much like any commercial advertisement. His method was effective; he received 15% more media coverage than Clinton.

First, there was Trump's highly inflammatory slogan, "Make America Great Again." This slogan is practically universally known; it's like a product slogan. However, Trump didn't choose to promote his slogan on television, but instead utilized other tools and newer technologies. Of course, both sides invested heavily in traditional media, but these channels seemed to have limited impact on the election. Hillary Clinton spent \$258 million on television advertising, while Trump only spent \$100 million. In cable television, Trump's market share was less than 1%. In digital advertising, Clinton also dominated, holding approximately 70%, but Trump avoided traditional political advertising channels, choosing a new platform—Twitter.

On Twitter, Trump's team used AI technology, political bots, and big data technology, achieving significant results. The propaganda on Twitter is so effective that it may be why his political ally, Elon Musk, bought the platform before the 2024 election.

Why are Twitter, big data, and bots, among other next-generation technologies, so effective? Perhaps it's due to the "megaphone effect." Simply put, these machines win by sheer numbers. If one account posts a sensational message, thousands of bot accounts will swarm it. For example, during the Pizzagate scandal, numerous accounts spread related memes and comments, and much of this traffic

appeared to originate outside the United States. These bot accounts' statements and actions are then retweeted by humans, creating overlap and a remarkable effect. Furthermore, Twitter's recommendation algorithm is attention-based, which is inherently suited to Trump's inflammatory and controversial content, and big data can provide precise user profiles for highly targeted political advertising.

We believe this is a combined effect of new platforms and technologies. If *Amusing Ourselves to Death* tells us that television technology replaced newspaper technology in the political arena, then platforms represented by Twitter, AI, and big data are the new "television technology." Hillary Clinton's team represents traditional information and communication technology; their political advertising on television was questionable in its effectiveness. Trump's victory was not only a victory of campaign strategy but also a victory of the new generation of media and communication technologies.

3.2 Technological War Between China and the USA

Regarding international politics, because people fully understand the influence of AI technology on domestic politics, every country must be cautious about such a force. We know that even Italy, as one of the dominant Western societies, initially banned ChatGPT. We will use the game between China and the US as a case study.

First, we focus on TikTok. It's not just a simple short video platform, but also a highly efficient propaganda platform. Both Democrats and Republicans are well aware of the influence of this new type of short video platform. The US has 150 million TikTok users, and they understand the implications of cultural soft power and platform ownership. This makes the US extremely fearful of Chinese manipulation. Although TikTok is headquartered in Singapore and operates independently from the Chinese Douyin Group, they still chose to buy TikTok through legislative procedures.[\[Kei25\]](#) This is certainly not a simple market behavior, but a struggle for data sovereignty.

Meanwhile, the US is pursuing a policy of restrictive border control, imposing export controls on high-end GPUs, advanced process chips, quantum computing, and other fields (small yard, high fence). It also refuses to admit Chinese students in key technology areas and even bans Chinese students from entering the US.[\[Cau25\]](#) This forces the world to split from a unified large market into smaller regions based on political alliances. For China, domestic substitution has always been a national strategy. All Chinese people have heard the term "chokehold," meaning that if China cannot master key technologies or production processes, it will be manipulated by hostile countries. This confrontation is actually a manifestation of the nascent protectionism stemming from digital sovereignty.

This seems paradoxical. Shouldn't technological progress promote globalization? In reality, globalization has never been entirely determined by technology. Rather, shifts in technological power do alter a nation's attitude towards and level of participation in globalization. While technology certainly possesses the potential to connect the world, its profound reshaping of power forces major powers to firmly grasp dominant technologies to ensure their political independence.

4 Conclusion

In summary, the development of ICTs and data science is far more than just technological progress; it profoundly impacts power structures and the patterns and objectives of competition between powers. It reshapes political competition at the micro level and the global political and economic landscape at the macro level. Whoever controls technology controls the future.

At the domestic political level, from the printing press to television advertising, and now to competition in big data technology...In marketing, we see those who possess the latest technology gaining power. Technology has propelled the marketization of politics, while simultaneously highlighting the importance of technological means.

Furthermore, because technological means have a disruptive impact on domestic politics that surpasses traditional methods, the importance of digital sovereignty has become apparent. When major powers no longer hold absolute technological superiority, they must intervene in the market and implement technological protectionism to ensure domestic political stability. These technologies thus influence domestic politics and further impact the international situation; whoever controls the technology controls the future.

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